

Deepak Sathyan

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📍 Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, College Station, TX

Education

University of Maryland

Ph.D. in Physics

Thesis: Unifying Searches for New Physics with Precision Measurements of the W Boson Mass

Advisor: Dr. Kaustubh Agashe

College Park, MD

Aug 2018 – July 2024

Boston University

Bachelor of Arts in Physics, Summa Cum Laude with Honors

GPA: 3.9

Minor: Mathematics

Thesis: Muon ($g-2$): Searching for the Muon Electric Dipole Moment

Advisor: Dr. Robert Carey

Boston, MA

Aug 2014 – May 2018

Experience

Postdoctoral Research Associate

Mitchell Institute for Fundamental Physics and Astronomy
Texas A&M University

College Station, TX

Sept 2024 – Aug 2027

3 years

Research Interests

Beyond the Standard Model searches with Collider Physics

searches at the LHC, sub-GeV dark matter, invisible new physics via precision measurements, Kaluza-Klein modes in Randall-Sundrum models, baryon number violation models, muon collider phenomenology

Searches for sub-GeV Dark Matter

Constraints on dark matter-neutrino interactions, dark photon production at the LHC, dark sector production at muon collider, astrophysical signals of cosmic ray-dark matter scattering, astrophysical sources of sub-GeV dark matter

Neutrino Physics

Supernova neutrino attenuation and detection, neutrino oscillations in open systems, neutrinophilic mediators

Articles in Review

3. Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering

B. Dutta, D. Goswami, J. Kumar, M. Rai, D. Sathyan

arxiv.org/abs/2605.08010

May 2026

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| <p>2. A Baryon and Lepton Number Violation Model Testable at the LHC
 A. Bhoonah, F. Burk, D. Liu, T. Ou, D. Sathyan
 arxiv.org/abs/2508.21064</p> | <p>Aug 2025</p> |
| <p>1. New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis
 P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
 arxiv.org/abs/2507.01000</p> | <p>July 2025</p> |

Articles in Refereed Journals

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| <p>5. New laboratory constraints on neutrinophilic mediators
 P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
 arxiv.org/abs/2407.12738 (Phys.Lett.B 868 (2025) 139765)</p> | <p>July 2024</p> |
| <p>4. ‘Unification’ of BSM searches and SM measurements: the case of lepton+MET and m_W
 K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
 arxiv.org/abs/2404.17574 (JHEP 02 (2025) 139)</p> | <p>Apr 2024</p> |
| <p>3. A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET
 K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
 arxiv.org/abs/2310.13687 (Phys.Lett.B 855 (2024) 138774)</p> | <p>Oct 2023</p> |
| <p>2. Energy-peak based method to measure top quark mass via B-hadron decay lengths
 K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, D. Sathyan
 arxiv.org/abs/2212.03929 (JHEP 06 (2023) 021)</p> | <p>Dec 2022</p> |
| <p>1. LHC Signals for KK Graviton from an Extended Warped Extra Dimension
 K. Agashe, M. Ekhaterachian, D. Kim, D. Sathyan
 arxiv.org/abs/2008.06480 (JHEP 11 (2020) 109)</p> | <p>Aug 2020</p> |

Articles in Preparation

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| <p>1. LHC as a Beam Dump: Probing Light Dark Photons
 B. Dutta, A. Karthikeyan, D. Kim, H. Kim, T. Kim, D. Sathyan
 arxiv.org/a/sathyan_d_1</p> | <p>Sept 2026</p> |
| <p>2. New Probes of Dark Sectors at a Muon Collider
 B. Dutta, A. Karthikeyan, D. Kim, K. Kelly, D. Sathyan
 arxiv.org/a/sathyan_d_1</p> | <p>Jan 2027</p> |
| <p>3. Exploring open systems approach to neutrino oscillations
 A. Chandra Shekar, K. Kelly, D. Sathyan, L. Strigari, T. Zhou
 arxiv.org/a/sathyan_d_1</p> | <p>Mar 2027</p> |

4. **LHC Signals for KK Gravitons Producing 8-Particle Final States** Mar 2027
K. Agashe, D. Kim, P. Maksimovic, S. Mondal, M. Osherson, K. Panchal, D. Plotnikov, D. Sathyan
arxiv.org/a/sathyan_d_1

Ph.D. Thesis

1. **Unifying Searches for New Physics with Precision Measurements of the W Boson Mass** 2024
D. Sathyan
drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9 (Ph.D. Thesis)

Snowmass2021 Contributions

5. **TF07 Snowmass Report: Theory of Collider Phenomena** Oct 2022
F. Maltoni et al.
arxiv.org/abs/2210.02591
4. **Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021** Sept 2022
K. Agashe et al.
arxiv.org/abs/2209.11267
3. **Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021** Sept 2022
T. Bose et al.
arxiv.org/abs/2209.13128
2. **Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond** Apr 2022
K. Agashe, S. Airen, R. Franceschini, D. Kim, D. Sathyan
arxiv.org/abs/2204.02928
1. **Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models** Mar 2022
K. Agashe, J.H. Collins, P. Du, M. Ekhaterachian, S. Hong, D. Kim, R.K. Mishra, D. Sathyan
arxiv.org/abs/2203.13305

Muon g-2 Articles

3. **The straw tracking detector for the Fermilab Muon g-2 Experiment** Nov 2021
B.T. King et al.
arxiv.org/abs/2111.02076 (JINST 17 (2022) P02035)
2. **Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab** Apr 2021
Muon g-2 Collaboration
arxiv.org/abs/2104.03240 (Phys.Rev.Accel.Beams 24 (2021) 044002)

1. Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm

Apr 2021

Muon g-2 Collaboration

arxiv.org/abs/2104.03281 (Phys.Rev.Lett. 126 (2021) 141801)

Seminars

7. CMS Exotica general meeting, May 2024

Unification of Searches and Measurements: Probing BSM with the W boson mass measurement

6. Seminar at Texas A&M, December 2023

A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

5. Seminar at UT Austin, December 2023

A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

4. CMS SUSY general meeting, November 2023

A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

3. HEP Seminar at University of Pittsburgh, November 2023

A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

2. HEP Seminar at Northwestern University, October 2023

A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

1. Theory Seminar at Washington University in St. Louis, October 2023

A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

Conference Talks

14. Light Dark World 2026 at Carleton University, July 2026

Can the LHC be sensitive to the Light Dark World?

13. CETUP* Workshop, June 2026

Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering

12. Particle Physics on the Plains at the University of Kansas, November 2025

A Baryon and Lepton Number Violation Model Testable at the LHC

11. CETUP* Workshop, June 2025

Can the LHC be sensitive to light dark mediators?

10. Phenomenology Conference at the University of Pittsburgh, May 2025

Can the LHC be sensitive to light dark mediators?

9. **Particle Physics on the Plains at the University of Kansas, November 2024**
A comprehensive analysis of supernova neutrino-dark matter interactions
8. **Texas TACOS @ UT Austin, October 2024**
A comprehensive analysis of supernova neutrino-dark matter interactions
7. **NuFact Conference, September 2024**
A comprehensive analysis of supernova neutrino-dark matter interactions
6. **Mitchell Conference at Texas A&M University, May 2024**
A comprehensive analysis of supernova neutrino-dark matter interactions
5. **Particle Physics on the Plains at the University of Kansas, October 2023**
A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
4. **Phenomenology Conference at the University of Pittsburgh, May 2023**
Probing Dark Matter-Neutrino Interactions via Supernova Neutrinos
3. **Phenomenology Conference at the University of Pittsburgh, May 2022**
Model-Independent Measurement of Top Quark Mass Using B-Hadron Decay Lengths (Part I)
2. **April APS Meeting, April 2021**
Signals of KK graviton from extended warped extra dimensions at the LHC (II)
1. **Phenomenology Conference at the University of Pittsburgh, May 2020**
Signals of KK graviton from extended warped extra dimensions at the LHC (II)

Conferences Organized

The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2026
Texas A&M University, May 27-30, 2026

The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2025
Texas A&M University, May 13-16, 2025

Awards

Breakthrough Prize in Fundamental Physics
Muon g-2 Collaboration

Apr 2026

Schools Attended

GGI Lectures on the Theory of Fundamental Interactions

Jan 2023 – Jan 2023

TASI 2024: The Frontiers of Particle Theory

June 2024 – June 2024

Outreach

BU Student Mentorship Program PRISM (2016-2018)

Mentored incoming freshman undergraduates in physics, providing guidance on courses, research opportunities, and resources

TAMU Physics and Engineering Festival (2025-2027)

Presented demonstrations of fluid dynamics to the general public, including high school students and families

University of South Dakota talk: "My path through particle physics research so far" (December 2024)

Presented my path through particle physics research to students, showing a variety of interesting ideas to pursue

Skills

Programming:

Mathematica, Python, C++

Physics Software:

MadGraph5_aMC@NLO, Pythia6, Pythia8, Delphes, ROOT, GEANT4, FeynRules